



# FloodWay

AI-Powered Flood Response Platform

**Predict • Verify • Guide**



**MAS**

**FIVE42**

# When flooding warnings arrive too late

In flood emergencies, every minute matters.

Yet across ASEAN, citizens still **depend on delayed alerts, fragmented communication, and unreliable information** during evacuation.

Floodway was created to transform flood response into real-time survival intelligence.





# Why Current Systems Fail?

## PROBLEM 1

### Reactive Warnings

*Alerts arrive after flooding begins.*

## PROBLEM 2

### Misinformation

*Communities rely on WhatsApp & social media.*

## PROBLEM 3

### No Personalized Guidance

*Citizens do not know which routes remain safe.*

## ASEAN Flood Impact :

**\$6B USD**

Worth of Infrastructure Damages

**250K**

Homes Affected

**1,200+**

People Died

# Solutions

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An illustration of a hand holding a glowing orange lightbulb with a white checkmark inside it. The background is decorated with colorful geometric shapes like circles, squares, and triangles in shades of green, blue, and purple.

1

## **AI Flood Prediction**

Machine learning predicts flood risk and danger zones.

2

## **Gemini Truth Verification**

AI verifies community-submitted flood reports before generating public alerts.

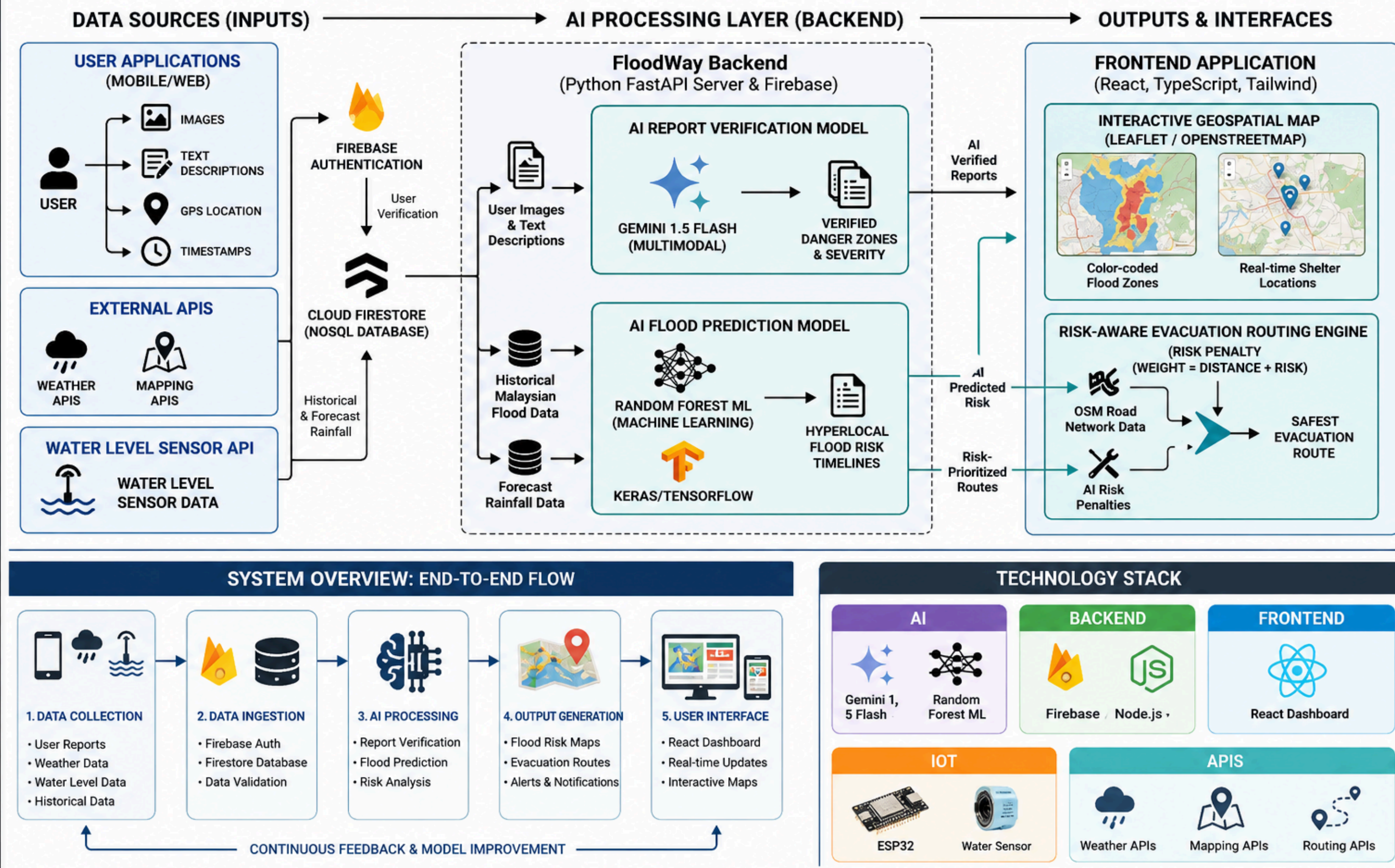
3

## **Intelligent Evacuation Routing**

Dynamic routing system generates safer evacuation path during emergencies.



# FloodWay: AI-Powered System Architecture and Data Flow Diagram







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**LIVE DEMO**



# Feasibility of Study

## EXISTING INFRASTRUCTURE

Integrates existing infrastructure instead of requiring entirely new deployments such as :

- river monitoring systems
- environmental datasets
- live sensors

## SCALABLE CLOUD ARCHITECTURE

Uses a modular cloud-based backend that supports regional scalling.

## REAL-TIME AI + IOT PROCESSING

Combines live sensor streaming, Random Forest flood prediction, Gemini verification and dynamic evacuation routing into a unifies disaster intelligence workflow.

# Winning Points

| Feature                           | Floodway | MyPublicInfoBanjir |
|-----------------------------------|----------|--------------------|
| AI + IOT Hybrid Prediction        | ✓        | ✗                  |
| Real-Time Safe Evacuation Routing | ✓        | ✗                  |
| AI Misinformation Verification    | ✓        | ✗                  |
| Personalized Flood Intelligence   | ✓        | ✗                  |
| ASEAN Scale Deployment            | ✓        | ✗                  |





# Market Opportunity

| GLOBAL                                                         | MALAYSIA                                              | OUR ADVANTAGE                                      | GO-TO-MARKET                  |
|----------------------------------------------------------------|-------------------------------------------------------|----------------------------------------------------|-------------------------------|
| 40M+ people affected by floods annually across ASEAN           | 250,000+ displaced during major floods                | AI + IoT Hybrid Prediction                         | Malaysia Pilot Deployment     |
| Smart disaster tech market projected to exceed USD18B+ by 2030 | Flood damages exceed RM6B+ annually                   | Real-Time Safe Evacuation Routing                  | Government & NGO Partnerships |
| Climate-driven floods increasing across Southeast Asia         | Flood-prone states include Kelantan, Selangor & Johor | Gemini AI Misinformation Verification              | Enterprise API Integration    |
| ASEAN countries require scalable disaster-response systems     | Existing systems remain reactive & fragmented         | Compatible with Existing Government Infrastructure | ASEAN Regional Expansion      |

# ASEAN Scalability

## Why ASEAN ?

- *Moonsoon Climate*
- *Urban Flooding*
- *Dense Populations*
- *Fragmented Emergency Communication*

SDG



## SCALABILITY ROADMAP

**Phase 1** – Malaysia Pilot

**Phase 2** – Indonesia & Thailand Expansion

**Phase 3** – ASEAN Disaster Intelligence Network

**Phase 4** – Advanced Predictive Disaster Analytics





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Team Five42™

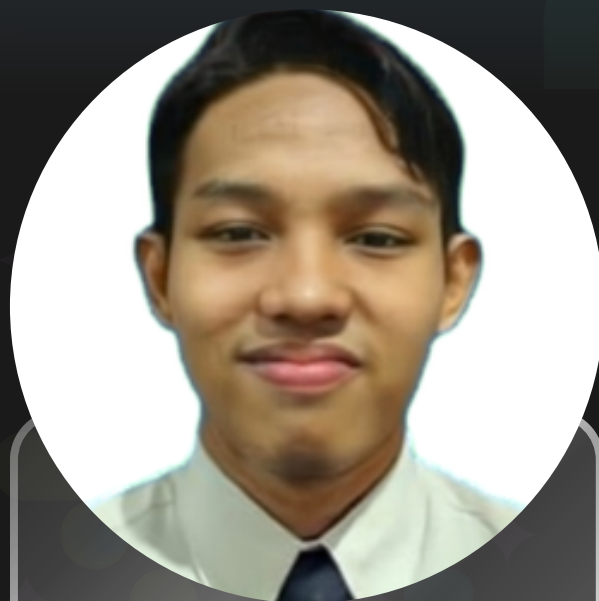
**We cannot stop the rain**

but we can stop people from being trapped.



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# Our Management Team



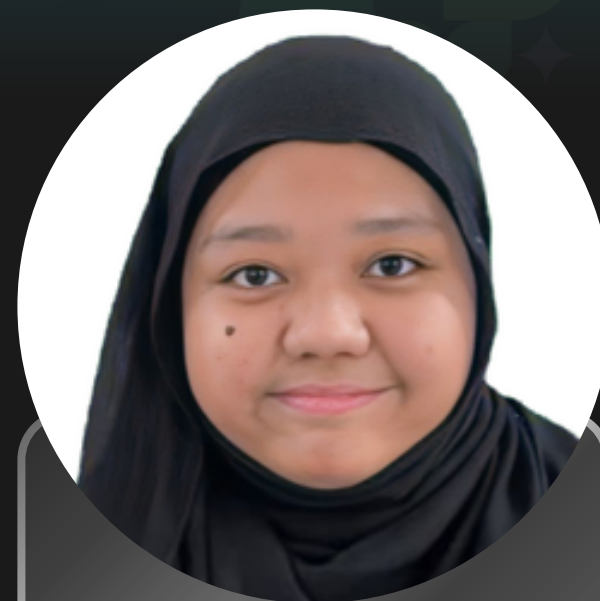
Muhammad  
Luqman Nurhakim

*Group Leader*



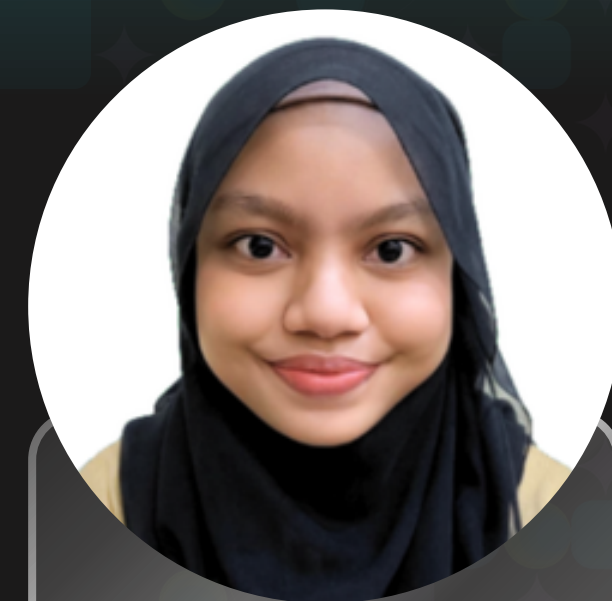
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